



## part III

# the glorious revival

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# arthur bailey

After the Second World War the first public challenge to the anti-Bayesian status quo came not from the military or university mathematicians and statisticians but from a Bible-quoting business executive named Arthur L. Bailey.

Bailey was an insurance actuary whose father had been fired and black-balled by every bank in Boston for telling his employers they should not be lending large sums of money to local politicians. So ostracized was the family that even Arthur's schoolmates stopped inviting him and his sister to parties. Turning his back on the New England establishment, Bailey enrolled at the University of Michigan in Ann Arbor. There he studied statistics in the mathematics department's actuarial program, earned a bachelor of science degree in 1928, and met his wife, Helen, who became an actuary for John Hancock Mutual Life before their children were born.<sup>1</sup>

Bailey's first job was, he liked to say, "in bananas," that is, in the statistics department of the United Fruit Company headquarters in Boston. When the department was eliminated during the Depression, Bailey wound up driving a fruit truck and chasing escaped tarantulas down Boston streets. He was lucky to have the job, and his family never lacked for bananas and oranges.

In 1937, after nine years in bananas, Bailey got a job in an unrelated field in New York City. There he was in charge of setting premium rates to cover risks involving automobiles, aircraft, manufacturing, burglary, and theft for the American Mutual Alliance, a consortium of mutual insurance companies.

Preferring church and community connections to the fair-weather friends of his youth, Bailey hid his growing professional success by living quietly in unpretentious New York suburbs. He relaxed by gardening, hiking

with his four children, and annotating a copy of Gray's *Botany* with the locations of his favorite wild orchids. His motto was, "Some people live in the past, some people live in the future, but the wisest ones live in the present."

Settling into his new job, Bailey was horrified to see "hard-shelled underwriters" using the semi-empirical, "sledge hammer" Bayesian techniques developed in 1918 for workers' compensation insurance.<sup>2</sup> University statisticians had long since virtually outlawed those methods, but as practical business people, actuaries refused to discard their prior knowledge and continued to modify their old data with new. Thus they based next year's premiums on this year's rates as refined and modified with new claims information. They did not ask what the new rates should be. Instead, they asked, "How much should the present rates be changed?" A Bayesian estimating how much ice cream someone would eat in the coming year, for example, would combine data about the individual's recent ice cream consumption with other information, such as national dessert trends.

As a modern statistical sophisticate, Bailey was scandalized. His professors, influenced by Ronald Fisher and Jerzy Neyman, had taught him that Bayesian priors were "more horrid than 'spit,'" in the words of a particularly polite actuary.<sup>3</sup> Statisticians should have no prior opinions about their next experiments or observations and should employ only directly relevant observations while rejecting peripheral, nonstatistical information. No standard methods even existed for evaluating the credibility of prior knowledge (about previous rates, for example) or for correlating it with additional statistical information.

Bailey spent his first year in New York trying to prove to himself that "all of the fancy actuarial [Bayesian] procedures of the casualty business were mathematically unsound."<sup>4</sup> After a year of intense mental struggle, however, he realized to his consternation that actuarial sledgehammering worked. He even preferred it to the elegance of frequentism. He positively liked formulae that described "actual data. . . . I realized that the hard-shelled underwriters were recognizing certain facts of life neglected by the statistical theorists."<sup>5</sup> He wanted to give more weight to a large volume of data than to the frequentists' small sample; doing so felt surprisingly "logical and reasonable." He concluded that only a "suicidal" actuary would use Fisher's method of maximum likelihood, which assigned a zero probability to nonevents.<sup>6</sup> Since many businesses file no insurance claims at all, Fisher's method would produce premiums too low to cover future losses.

Abandoning his initial suspicions of Bayes' rule, Bailey spent the Second

World War studying the problem. He worked alone, isolated from academic thinkers and from his actuarial colleagues, who scratched their heads at Bailey's brilliance.

After the war, in 1947, Bailey moved to the New York State Insurance Department as the regulatory agency's chief actuary. An insurance executive called him "the keeper of our consciences." As his colleagues boozed in hotel bars at conferences, Bailey sipped soft drinks and quoted occasionally from the Bible. During slack times, he read it. Some actuaries said "all manner of nasty things about Arthur Bailey," the executive continued, "but we learned to respect his integrity and stature from knowing him in the after-hours."

Bailey began writing an article summarizing his tumultuous change in attitude toward Bayes' rule. Although his old-fashioned notation was difficult to understand, he was building a mathematical foundation to justify the use of current rates as the priors in Bayes' theorem. He started his paper with a biblical justification for using prior beliefs: "If thou canst believe," he wrote, quoting the apostle Mark, "all things are possible to him that believeth." Then, reviewing Albert Whitney's mathematics for workers' compensation, Bailey affirmed the Bayesian roots of the Credibility theory developed for workers' compensation insurance years before. Credibility was central to actuarial thought, and while relative frequencies were relevant, so were other kinds of information. Bailey worked out mathematical methods for melding every scrap of available information into the initial body of data. He particularly tried to understand how to assign partial weights to supplementary evidence according to its credibility, that is, its subjective believability. His mathematical techniques would help actuaries systematically and consistently integrate thousands of old and new rates for different kinds of employers, activities, and locales. His working library included a 1940 reprint of Bayes' articles with a preface by Bell Telephone's Edward Molina. Like Molina, Bailey used Laplace's more complex and precise system instead of Thomas Bayes'.

By 1950 Bailey was a vice president of the Kemper Insurance Group in Chicago and a frequent after-dinner speaker at black-tie banquets of the Casualty Actuarial Society. He read his most famous paper on May 22, 1950. Its title explained a lot: "Credibility Procedures: Laplace's Generalization of Bayes' Rule and the Combination of Collateral [that is, prior] Knowledge with Observed Data."

For actuaries who could concentrate on a long scholarly paper after a heavy (and no doubt alcoholic) meal, Bailey's message must have been thrilling. First, he praised his colleagues for standing almost alone against

the statistical establishment and for staging the only organized revolt against the frequentists' sampling philosophy. Insurance statisticians marched "a step ahead" of others. Actuarial practice was an obscure and profound mystery, and it went "beyond anything that has been proven mathematically." But, he declared triumphantly, "it works. . . . They have made this demonstration many times. It does work!"<sup>8</sup>

Then he announced the startling news that their beloved Credibility formula was derived from Bayes' theorem. Practical actuaries had thought of Bayes as an abstract, temporal solution treating time sequences of priors and posteriors. But Bailey reminded his colleagues that Bayes' friend and editor, Richard Price, would be considered today an actuary. And he turned Bayes' imaginary table into a frontal attack on frequentists and the contentious Fisher. He concluded with a rousing call to reinstate prior knowledge in statistical theory. His challenge would occupy academic theorists for years. It was a fighting speech. Reading it later, Professor Richard von Mises of Harvard praised it wholeheartedly. Von Mises wrote Bailey that he hoped his speech would make "the unjustified and unreasonable attacks on the Bayes theory, initiated by R. A. Fisher, fade out."<sup>9</sup>

Unfortunately, Bailey did not live long to campaign for Bayes' rule. Four years after giving his most important speech, he suffered a heart attack at the age of 49 and died on August 12, 1954. His son blamed the fact that Bailey had started smoking in college and been unable to stop.

Still, a few practicing actuaries understood his message. The year of Bailey's death, one of his admirers was sipping a martini at the Insurance Company of North America's Christmas party when INA's chief executive officer, dressed as Santa Claus, asked an unthinkable question: Could anyone predict the probability of two planes colliding in midair?

Santa was asking his chief actuary, L. H. Longley-Cook, to make a prediction based on no experience at all. There had never been a serious midair collision of commercial planes. Without any past experience or repetitive experimentation, any orthodox statistician had to answer Santa's question with a resounding no. But the very British Longley-Cook stalled for time. "I really don't like these things mixed with martinis," he drawled. Nevertheless, the question gnawed at him. Within a year more Americans would be traveling by air than by railroad. Meanwhile, some statisticians were wondering if they could avoid using the ever-controversial subjective priors by making predictions based on no prior information at all.

Longley-Cook spent the holidays mulling over the problem, and on

January 6, 1955, he sent the CEO a prescient warning. Despite the industry's safety record, the available data on airline accidents in general made him expect "anything from 0 to 4 air carrier-to-air carrier collisions over the next ten years." In short, the company should prepare for a costly catastrophe by raising premium rates for air carriers and purchasing reinsurance. Two years later his prediction proved correct. A DC-7 and a Constellation collided over the Grand Canyon, killing 128 people in what was then commercial aviation's worst accident. Four years after that, a DC-8 jet and a Constellation collided over New York City, killing 133 people in the planes and in the apartments below.<sup>10</sup>

Later, Arthur Bailey's son, Robert A. Bailey, used Bayesian techniques to justify offering merit rates to good drivers. Motor vehicle casualty rates soared so high during the 1960s that half the Americans then alive could expect to be injured in a car accident during their lifetimes. Americans were buying more cars and driving more miles, but laws had not kept pace. There was no uniform road signage; most drivers and vehicles were tested or inspected only once in their lifetimes, if at all; penalties for drunk driving were light; and cars were designed without safety in mind. Insurers suffered heavy losses. A direct, up-front system to reward good drivers was needed, but merit rating was regarded as unsound because a single car had inadequate credibility. Using Bayes' rule, Robert Bailey and Leroy J. Simon showed that relevant data from Canada's safe-driving discounts could be used to update existing U.S. statistics.

Robert Bailey also used Bayesian procedures to rate insurance companies themselves by incorporating nonstatistical, subjective information such as opinions about a company's ownership, including the quality and drinking habits of its managers. In time, the insurance industry accumulated such enormous amounts of data that Bayes' rule, like the slide rule, became obsolete.

To the few actuaries who understood Arthur Bailey's work, he was a da Vinci or a Michelangelo: he had led their profession out of its dark ages.<sup>11</sup> News of his achievement percolated slowly and haphazardly to university theorists. During the early 1960s an actuarial professor at the University of Michigan, Allen L. Mayerson, wrote about Bailey's seminal role in Credibility theory. Professor of statistics Jimmie Savage, a new convert to Bayesian methods, was working in Ann Arbor at the time and later visited Bruno de Finetti, the Bayesian actuarial professor, at his vacation home on an island off Italy. The two attended a conference together in Trieste, where the Italian

spread the word about Bailey and the Bayesian origin of insurance Credibility. It was the first time most statisticians had heard of him.

Hans Bühlmann, who became a mathematics professor and president of ETH Zurich, remembers the excitement of that conference. He had spent a leave of absence studying in Neyman's statistics department in Berkeley in the 1950s, "when it was kind of dangerous to pronounce the Bayesian point of view." Taking up Bailey's challenge, Bühlmann produced a general Bayesian theory of credibility, which statisticians carried far beyond the world of actuaries and insurance. Carefully renaming the prior a "structural function," Bühlmann believed he helped Continental Europe escape some of the "religious" quarrels over Bayes' rule, quarrels that lay ahead for Anglo-Americans.<sup>12</sup>

# from tool to theology

While Arthur Bailey was transforming the sledgehammer of Credibility into Bayes' rule for the insurance industry, a postwar boom in statistics was elevating the method's lowly status. Gradually, Bayes would shed its reputation as a mere tool for solving practical problems and emerge in glorious Technicolor as an all-encompassing philosophy. Some would even call it a theology.

The Second World War had radically upgraded the stature, financial prospects, and career opportunities of applied mathematicians in the United States. The military was profoundly impressed by its wartime experience with statistics and operations research, and during the late 1940s the government poured money into science and statistics. Military funding officers roamed university hallways trying to persuade often reluctant statisticians to apply for grants. Naval leaders, convinced that postwar science needed a jump-start to prime technology's pump, organized the Office of Naval Research, the first federal agency formed expressly to finance scientific research. Until the National Science Foundation was created in 1950, the U.S. Navy supported much of the nation's mathematical and statistical research, whether classified or unclassified, basic or applied. Other funding came from the U.S. Army, the U.S. Air Force, and the National Institutes of Health.

A generation of pure mathematicians who had made exciting, life-or-death decisions during the war soon switched to applied mathematics and statistics. As the statistics capital of the world moved from Britain to the United States, the field exploded. Amid such spectacular growth, the number of theoretical statisticians increased a hundred-fold. Settling into mathematics departments, they coined new terms like "mathematical statistics" and "theoretical statistics."

In these boom times, even Bayesians could get jobs in elite research institutions. At one end of the Bayesian spectrum was a small band of evangelists intent on making their theories mathematically and academically respectable. At the other end were practitioners who wanted to play key roles in science instead of in formalistic mathematical exercises.

In the face of jangling changes and new attitudes, the wartime marriage of convenience between abstract and applied mathematicians fell apart. Statisticians complained that pure mathematicians regarded useful research as “something for peasants,” akin to washing dishes and sweeping streets.<sup>1</sup> Jack Good claimed the mathematicians at Virginia Tech, home of the nation’s third largest statistics department in the 1960s, loathed problem solvers.<sup>2</sup>

Frisky with federal funds, statisticians and data analysts divorced themselves from mathematics departments and formed their own enclaves. Yet even there tension sizzled between abstract theorizing and scientific applications, albeit in more decorous privacy. Serial schisms continue to this day, with applied mathematicians occupying—depending on the university—departments of mathematics, applied mathematics, statistics, biostatistics, and computer science.

Jerzy Neyman’s laboratory at Berkeley, then the largest and most important statistics center in the world, developed fundamental sampling theories and reigned over this fractious profession for years after the Second World War. But Neyman’s laboratory developed fissures of its own. Unable to compete with the soaring demand for statisticians, the department hired and promoted its own students and became ingrown. When a student tried to solve a blackboard problem unconventionally, Neyman grabbed his hand and forced it to write the answer Neyman’s way. For 40 years most of his hires were frequentists, and outsiders called the group “Jesus and his disciples.”<sup>3</sup> Neyman continued to run his institute into his 80s.

Although both were fervent anti-Bayesians, Neyman and Fisher battled to the end, neither willing to admit that the other might be using the technique that best fit his own needs. For Fisher, the stakes were high: “We are quite in danger of sending highly trained and intelligent young men out into the world with tables of erroneous numbers under their arms, and with a dense fog in the place where their brains ought to be. In this century, of course, they will be working on guided missiles and advising the medical profession on the control of diseases, and there is no limit to the extent to which they could impede every sort of national effort.”<sup>4</sup> He described Ney-

man as “some hundred years out of date, . . . partly incapacitated by the crooked reasoning.”<sup>5</sup> Neyman called Fisher’s researches “insidious because, in a skillfully hidden manner, they involve unjustified claims of priority.”<sup>6</sup> And so it went. At the age of 85, Neyman declared loftily, “[Bayes] does not interest me. I am interested in frequencies.”<sup>7</sup>

To Bayesian sympathizers, frequentism began to look like a Rube Goldberg cartoon of loosely connected ad hoceries, tests, and procedures that arose independently instead of growing in a unified, logical manner out of probability. The joke was that if you didn’t like the result of your frequentist analysis, you just redid it using a different test. By comparison, Bayes’ rule seemed to have an overall rationale. As the number of statisticians, symposia, articles, and journals multiplied, a series of publications issued around 1950 began to attract attention to the heretofore invisible world of Bayes’ rule.

Bayes stood poised for another of its periodic rebirths as three mathematicians, Jack Good, Leonard Jimmie Savage, and Dennis V. Lindley, tackled the job of turning Bayes’ rule into a respectable form of mathematics and a logical, coherent methodology. The first publication heralding the Bayesian revival was a book by Good, Alan Turing’s wartime assistant. As Good explained, “After the war, he [Turing] didn’t have time to write about statistics because he was too busy designing computers and computer languages, and speculating about artificial intelligence and the chemical basis of morphogenesis, so with his permission, I developed his idea . . . in considerable detail.”<sup>8</sup> Good finished the first draft of *Probability and the Weighing of Evidence* in 1946 but could not get it published until 1950, the same year Arthur Bailey issued his Bayesian manifesto for actuaries. Much of the delay, Good explained, was caused by the continuation of wartime secrecy during the Cold War.

At first, his book fell on deaf ears. He was unaccustomed to teaching or explaining his ideas, and no one knew he had used Bayes to help break the Enigma codes. When he gave a talk about his “neo-Bayesian or neo/Bayes-Laplace philosophy” at a Royal Statistical Society conference, his style was clipped, and he did not waste words.<sup>9</sup> Lindley, who was in the audience, reported, “He did not get his ideas across to us. We should have paid much more respect to what he was saying because he was way in advance of us in many ways.”<sup>10</sup>

After the war Good continued doing classified cryptography for the British government and frequently used equal priors to help decide what hypotheses he should follow up. When David Kahn’s bestseller *The Codebreak-*

ers was published in 1967, the National Security Agency censored a passage identifying Good as one of Britain's top three cryptanalysts. He was at the time one of the world's most knowledgeable people about the coding industry. Good was quick, smart, original, armed with a fabulous memory, and unconventional enough to think about the paranormal and astrology and to join Mensa, the organization for people with high IQs. He introduced himself with a handshake and the words, "I am Good."<sup>11</sup>

From the Second World War on, everything technical about cryptography was classified, and while Good obeyed the restraints, he chafed against them and looked for ways to evade censorship. To reveal an ultraclassified technique used by Turing to find pairs and triplets of letters indicating the German submariners' code of the day, Good wrote about a favorite British hobby, bird watching. What if, he suggested, an avid bird watcher spotted 180 different bird species? Many of them would be represented by only one bird; logically, the bird watcher must have entirely missed many other species. Counting those missing species as zero (as a frequentist would have done) has the deleterious effect of asserting that missing species can never be found. Turing decided to assign those missing species a tiny possibility, a probability that is not zero. He was trying to learn about rare letter groupings that did not appear in his collection of intercepted German messages. By estimating the frequency of missing species in his sample, he could use Bayes to estimate the probability of those letter groupings appearing in a much larger sample of messages and in the very next Enigma message he received. Decades later, DNA decoders and artificial intelligence analysts would adopt the same technique.

Clever as he was, Good could be difficult to get along with, and he moved from post to post. After he spent a year at a cryptography think tank, the Institute for Defense Analyses (then at Princeton University), many coworkers were relieved to see him go. In 1967 Good moved permanently to Virginia Polytechnic Institute and State University in Blacksburg. At his insistence, his contract stipulated that he would always be paid one dollar more than the football coach. He worked far from the Bayesian mainstream, however; during the 1960s Bayes' rule in the United States was concentrated at the universities of Chicago and Wisconsin and at Harvard and Carnegie Mellon.

Sidelined by geography and silenced by the British government's classification of his work with Turing, Good mailed unsolicited carbons of his typed curriculum vitae—what he called his Private List of more than 800 articles and four books<sup>12</sup>—to startled colleagues. He numbered every work

and marked a significant portion of them as classified. Only as the British slowly declassified his cryptanalysis work could he reveal Bayes' success with the Enigma code. When that happened, he bought a vanity license plate emblazoned with his James Bond spy status and his initials, 007 IJG.

Hampered by governmental secrecy, his own personality, and an inability to explain his work, Good remained an independent voice within the Bayesian community as two others became its intellectual leaders.

Unlike Good, Dennis Lindley and Jimmie Savage evolved almost accidentally as Bayesians. When Lindley was a boy during the German bombing of London, a remarkable mathematics teacher named M. P. Meshenberg tutored him in the school's air raid shelter. Meshenberg convinced Dennis's father, a roofer proud to have never read a book, that the boy should not quit school early or be apprenticed to an architect. Because of Meshenberg, Dennis stayed in school and won a mathematics scholarship to attend Cambridge University. Later in the war, when the British government asked mathematicians to learn some statistics, Lindley helped introduce statistical quality control and inspection into armaments production for the Ministry of Supply.

After the war he returned to Cambridge, the British center of probability, where Jeffreys, Fisher, Turing, and Good had either worked or studied. There Lindley became interested in turning the statisticians' collection of miscellaneous tools into a "respectable branch of mathematics," a complete body of thought based on axioms and proven theorems.<sup>13</sup> Andrei Kolmogorov had done the same for probability in general in the 1930s. Since Fisher in particular often arrived at his ideas intuitively and neglected mathematical details, there was plenty of room for another mathematician to straighten things out logically.

In 1954, a year after publishing a lengthy article summarizing his project, Lindley visited the University of Chicago, only to realize that Savage had done an even better job of it. Although Lindley and Savage would soon become leading spokesmen for Bayes' rule, neither realized at this point he was headed down a slippery slope toward Bayes. Each thought he had merely put traditional statistical techniques on a rigorous mathematical footing. Only later did they realize they could not move logically from their rigorous axioms and theorems to the ad hoc methods of frequentism. Lindley said, "We were both fools because we failed completely to recognize the consequences of what we were doing."<sup>14</sup>

Despite being almost blind, Savage was immensely learned on an ency-

clopedic range of topics. His father, a Jewish East European immigrant with a third-grade education, had changed the family's name from Ogushevitz to Savage and settled in Detroit. Both Jimmie and his brother Richard were born with extreme myopia and involuntary eye movements. As an adult, before crossing a street Jimmie would wait five or ten minutes to make sure there were no oncoming cars, and attending lectures he would approach the blackboard and peer at it through a powerful monocular. The brothers could read quite comfortably, however, and as children called themselves "reading machines";<sup>15</sup> their mother, a high school graduate and nurse, had kept them supplied with library books. Reading was always a privilege to be cherished, and Jimmie read with a rare intensity and developed the embarrassing habit of questioning everything. His wide-ranging studies and insatiable curiosity would alter the history of Bayes' rule.

Because of his eyesight, however, Savage almost missed getting a college education. His teachers considered him feeble-minded and unsuited for higher studies. He was finally admitted to Wayne (later Wayne State) University in Detroit. From there he transferred to the chemistry department at the University of Michigan, only to be rejected again, this time as unfit for laboratory work. A kindly mathematics professor, G. Y. Rainich, rescued him by teaching a class of visually impaired students in total darkness. Rainich called it "mental geometry . . . just like in Russia," where many schools could not afford candles.<sup>16</sup> Three students in the class, including Savage, earned doctorates.

During the Second World War Savage worked in the Statistical Research Group at Columbia University with the future Nobel Prize-winning economist Milton Friedman. The experience persuaded Savage to switch from pure mathematics to statistics. After the war he moved to the University of Chicago, a center of scientific excitement, thanks in large part to the dazzling Nobel Prize winner Enrico Fermi, the last physicist to excel at both experimentation and theory. Fermi himself used Bayes. In the autumn of 1953, when Jay Orear, one of Fermi's graduate students, was struggling with a problem involving three unknown quantities, Fermi told him to use a simple analytic method that he called Bayes' theorem and that he had derived from C. F. Gauss. A year later, when Fermi died at the age of 53, Bayes' rule lost a stellar supporter in the physical sciences.

Fermi was not the only important physicist to use Bayesian methods during this period. A few years later, at Cornell University, Richard Feynman suggested using Bayes' rule to compare contending theories in physics.

Feynman would later dramatize a Bayesian study by blaming rigid O-rings for the *Challenger* shuttle explosion.

During this exciting period in 1950s Chicago, Savage and Allen Wallis founded the university's statistics department, and Savage attracted a number of young stars in the field. Reading widely, Savage discovered the work of Émile Borel, Frank Ramsey, and Bruno de Finetti from the 1920s and 1930s legitimizing the subjectivity in Bayesian methods.

Savage's revolutionary book *Foundations of Statistics* was the third in the series of pathbreaking Bayesian publications in the fifties. It appeared in 1954, four years after Bailey's insurance paper and Good's book and one year after Lindley's paper. Because of Ramsey's early death, it fell to Savage to develop the young philosopher's ideas about utility and to turn Bayes' rule for making inferences based on observations into a tool for decision making and action.

Almost defiantly, Savage proclaimed himself a subjectivist and a personalist. Subjective probability was a measure of belief. It was something you were willing to use for a bet, particularly on a horse race, where bettors share the same information about a horse but come to different conclusions about its chances and where the race itself can never be precisely replicated. Subjective opinions and professional expertise about science, medicine, law, engineering, archaeology, and other fields were to be quantified and incorporated into statistical analyses.

More than anyone else Savage forced people to think about combining two concepts: utility (the quantification of reward) and probability (the quantification of uncertainty). He argued that rational people make subjective choices to minimize expected losses.

Savage was confronting the thorniest objection to Bayesian methods: "If prior opinions can differ from one researcher to the next, what happens to scientific objectivity in data analysis?"<sup>17</sup> Elaborating on Jeffreys, Savage answered as follows: as the amount of data increases, subjectivists move into agreement, the way scientists come to a consensus as evidence accumulates about, say, the greenhouse effect or about cigarettes being the leading cause of lung cancer. When they have little data, scientists disagree and are subjectivists; when they have piles of data, they agree and become objectivists. Lindley agreed: "That's the way science is done."<sup>18</sup>

But when Savage trumpeted the mathematical treatment of personal opinion, no one—not even he and Lindley—realized yet that he had written the Bayesian Bible. "Neither of us would have known at the time what was meant by saying we were Bayesians," Lindley said. Savage's book did not

use the term “Bayesian” at all and referred to Bayes’ rule only once. Savage’s views and his book gained popularity slowly, even among those predisposed to Bayes’ rule. Many had hoped for a how-to manual like Fisher’s *Statistical Methods for Research Workers*. Lacking computational machinery to implement their ideas, Bayesians were limited to a few simple problems involving easily solved integrals and would spend years adapting centuries-old methods for calculating them. Savage, though, said he was “little inclined to high speed machines for help. This is no doubt partly due to my being reactionary . . . but my main interests are in the qualitative. . . . Tables of functions depending on several parameters are almost unprintable and, when printed quite unintelligible.”<sup>19</sup> Savage continued instead to prove abstract mathematical theorems and work on building a logical foundation for Bayesian methods.

His applications were too whimsical to be useful: what is the probability that aspirin curls rabbits’ ears? what is the most probable speed of neon light through beer? Some thought Savage’s failure to tackle serious problems impeded the spread of Bayesian methods. Lindley complained, “Perhaps statistics would have benefited more if he had not been so punctilious in replying to correspondents and so helpful with students, and instead developed more operational methods that the writers and graduates could have used.”<sup>20</sup>

Some readers were also troubled by the fact that Savage used aspects of frequentism to argue for Bayes’ subjective priors, taboo since the nineteenth century. As Savage explained, when he wrote the book he was “not yet a personalistic Bayesian.” He thought he came to Bayesian statistics “seriously only through recognition of the likelihood principle; and it took me a year or two to make the transition.”<sup>21</sup>

According to the likelihood principle, all the information in experimental data gets encapsulated in the likelihood portion of Bayes’ theorem, the part describing the probability of objective new data; the prior played no role. Practically speaking, the principle greatly streamlined analysis. Scientists could stop running an experiment when they were satisfied with the result or ran out of time, money, and patience; non-Bayesians had to continue until some frequency criterion was met. Bayesians would also be able to concentrate on what happened, not on what could have happened according to Neyman-Pearson’s sampling plan.

The transition to Bayes took Savage several years, but by the early 1960s he had accepted its logic wholeheartedly, fusing subjective probability with new statistical tools for scientific inference and decision making. As far as Savage was concerned, Bayes’ rule filled a need that other statistical proce-

dures could not. Frequentism's origin in genetics and biology meant it was involved with group phenomena, populations, and large aggregations of similar objects. As for using statistical methods in biology or physics, the Nobel Prize-winning physicist Erwin Schrödinger said, "The individual case [is] entirely devoid of interest."<sup>22</sup> Bayesians like Savage, though, could work with isolated one-time events, such as the probability that a chair weighs 20 pounds, that a plane would be late, or that the United States would be at war in five years.

Bayesians could also combine information from different sources, treat observables as random variables, and assign probabilities to all of them, whether they formed a bell-shaped curve or some other shape. Bayesians used all their available data because each fact could change the answer by a small amount. Frequency-based statisticians threw up their hands when Savage inquired whimsically, "Does whiskey do more harm than good in the treatment of snake bite?" Bayesians grinned and retorted, "Whiskey probably does more harm than good."<sup>23</sup>

As a movement, Bayes was looking more akin to a philosophy—even a religion or a state of mind—than to a true-or-false scientific law like plate tectonics. According to David Spiegelhalter of Cambridge University, "It's much more basic. . . . A huge sway of scientists says you can't use probability to express your lack of knowledge or one-time events that don't have any frequency to it. Probability came very late into civilization. . . . [and many scientists find it] rather disturbing because it's not a process of discovery. It's more a process of interpretation."<sup>24</sup>

"Mathematical scientists often sense a combination of harmony and power in certain formulas," explains Robert E. Kass, a Bayesian at Carnegie Mellon University. "There is at once a deep esthetic experience and a pragmatic recognition of profound consequences, leading to what Einstein called 'the cosmic religious feeling.' Bayes Theorem gives such a feeling. It says there is a simple and elegant way to combine current information with prior experience in order to state how much is known. It implies that sufficiently good data will bring previously disparate observers to agreement. It makes full use of available information, and it produces decisions having the least possible error rate. Bayes' Theorem is awe-inspiring." Unfortunately, Kass continued, "when people are captivated by its spell, they tend to proselytize and become blinded to its fundamental vulnerability. . . . [that] its magical powers depend on the validity of its probabilistic inputs."<sup>25</sup>

With zealots proselytizing Bayes as an all-encompassing panacea, the

method inspired both religious devotion and dogmatic opposition. The battle between Bayesians and their equally fervent foes raged for decades and alienated many bystanders. As one onlooker reflected, "It was a huge food fight. It was devastating. They hated each other."<sup>26</sup> A prominent statistician lamented, "Bayesian statisticians do not stick closely enough to the pattern laid down by Bayes himself: if they would only do as he did and publish posthumously we should all be saved a lot of trouble."<sup>27</sup>

Savage became one of the believers. He developed into a full-blown, messianic Bayesian, "the most extreme advocate of a Bayesian . . . ever seen," William Kruskal of the University of Chicago said. Savage recast the controversy over Bayes' rule in its most extreme form as subjectivity versus objectivity. For him, as for Lindley, the rule was the one-and-only, winner-take-all method for reaching conclusions in the face of uncertainty. Bayes' rule was right and rational, they felt, and other views were wrong, and it was neither necessary nor desirable to admit compromise.

"Personal probability . . . became for [Savage] the only sensible approach to probability and statistics," Kruskal recalled sadly. "If one were not in substantial agreement with him, one was inimical, or stupid, or at the least inattentive to an important scientific development. This attitude, no doubt sharpened by personal difficulties and by the mordant rhetoric of some anti-Bayesians, exacerbated relationships between Jimmie Savage and many old professional friends."<sup>28</sup>

Savage's last year at Chicago, 1960, was fraught with turmoil. Although his department colleagues knew nothing about it, the administration was trying to abolish the statistics department, and Savage was fighting to get the decision reversed. His marriage was disintegrating and, hoping to save it, he moved to the University of Michigan. As he departed, he told his colleagues, "I proved the Bayesian argument in 1954. None of you have found a flaw in the proof and yet you still deny it. Why?"<sup>29</sup> When he tried to return to Chicago, members of the department he had formed and chaired voted against rehiring him. At first, no other American or British university would offer him a position. In 1964 he moved to Yale University, remarried, and achieved some level of tranquility.

In 1971, at the age of 53, Savage died suddenly of a heart attack. His death at midcareer deprived American Bayesians of their leading spokesman. The *New Haven Register* had another perspective. Savage had cowritten a book called *How to Gamble If You Must*. For Bayesians, all assumptions about the future were risky, and gambling was the paradigm of decision making.

The Register headlined his obituary, “Yale Statistician Leonard Savage Dies; Authored Book on Gambling.”

In the meantime, Lindley had moved back to Britain, where for many years he was the only Bayesian in a position of authority. In time he built not just Bayesian theory but also strong Bayesian research groups, first at the University College of Wales in Aberystwyth and then at University College London. The latter had England’s most important statistical department and was a temple of frequentism. When Lindley arrived, a colleague said it was “as though a Jehovah’s Witness had been elected Pope.”<sup>30</sup> Lindley complained that he “inherited” several statisticians who “would not change their view of statistics.”<sup>31</sup> He said, “The general attitude [was] to turn their heads the other way.”<sup>32</sup>

In an era when many sneered at Bayes, it took courage to create Europe’s leading Bayesian department. Often the only Bayesian at meetings of the Royal Statistical Society and certainly the only combative one, Lindley defended Bayes’ rule like a fearless terrier or a devil’s advocate. In return, he was tolerated almost as comic relief. “Bayesian statistics is not a branch of statistics,” he argued. “It is a way of looking at the whole of statistics.”

Lindley became known as a modern-age revolutionary. He fought to get Bayesians appointed, professorship by professorship, until the United Kingdom had a core of ten Bayesian departments. Eventually, Britain became more sympathetic to the method than the United States, where Neyman maintained Berkeley as an anti-Bayesian bunker. Still, the process left scars: despite Lindley’s landmark contributions he was never named a Fellow of the Royal Society. In 1977, at the age of 54, Lindley forsook the administrative chores he hated and retired early. He celebrated his freedom by growing a beard and becoming what he called “an itinerant scholar” for Bayes’ rule.<sup>33</sup>

Thanks to Lindley in Britain and Savage in the United States, Bayesian theory came of age in the 1960s. The philosophical rationale for using Bayesian methods had been largely settled. It was becoming the only mathematics of uncertainty with an explicit, powerful, and secure foundation in logic. How to apply it, though, remained a controversial question.

Lindley’s enormous influence as a teacher and organizer bore fruit in the generation to come, while Savage’s book spread Bayesian methods to the military and to business, history, game theory, psychology, and beyond. Although Savage wrote about rabbit ears and neon light in beer, he personally encouraged researchers who would apply Bayes’ rule to life-and-death problems.

# jerome cornfield, lung cancer, and heart attacks

Bayes came to medical research through the efforts of a single scientist, Jerome Cornfield, whose only degree was a B.A. in history and who relied on the rule to identify the causes of lung cancer and heart attacks.

Lung cancer, extremely rare before 1900 and still uncommon in 1930, sprang up as if out of nowhere shortly after the Second World War. By 1952 it was killing 321 people per million per year in England and Wales. A year later approximately 30,000 new cases were diagnosed in the United States. No other form of cancer showed such a catastrophic leap. Studies in Europe, Turkey, and Japan confirmed the puzzling plague. There seemed to be something special about the disease.

But what could it be? Its cause was unknown. Pathologists thought the increase in lung cancer might be due to improvements in diagnostic methods or to the natural aging of the population. Others blamed exhaust from factories or the growing number of automobiles, tar particles from modern asphalt pavements, or England's infamous smog from homes heated with open coal-burning fires. Cigarettes, mass-produced since the invention of a cigarette-making machine in 1880, had been patriotically shipped to soldiers during the First World War. Animal studies, though, had failed to demonstrate that tobacco tar was carcinogenic.

As early as 1937 a small-scale study in Germany had pointed ever so tentatively to cigarette smoke. But there were doubts about that too. While 80% of middle-aged men in England and Wales smoked cigarettes, tobacco consumption per capita had dropped slightly. And fumes from cigarettes, which had replaced cigars and pipes, did not seem worse than other smoke.

The world's most famous biostatistician, Austin Bradford "Tony" Hill, was intrigued. He called himself an arithmetician rather than a mathematician or statistician and, in a series of articles in *The Lancet*, used straightforward logic to persuade the medical community to objectively quantify its research findings. During the late 1940s, two decades after Ronald Fisher had introduced randomization to agricultural experimentation, Hill introduced randomization to medical research. Inaugurating the modern controlled clinical trial, Hill showed that pertussis vaccine reduced children's whooping cough cases by 78% and that streptomycin was effective against pulmonary tuberculosis. Bradford Hill became so famous that a letter addressed to "Lord Hill, Bradford, England" reached him.

To identify the most probable causes of the catastrophic increase in lung cancer, Hill and a young physician and epidemiologist, Richard Doll, organized interviews with patients with and without lung cancer in 20 London-area hospitals. All were questioned about their past activities and exposures. The results, published in 1950, were shockingly clear. Of 649 men with lung cancer, only 2 were nonsmokers; a high proportion of the lung cancer patients were heavy cigarette smokers, and their death rate was 20 times higher than that of nonsmokers. A large American study by Ernst L. Wynder and Evarts A. Graham confirmed the British result the same year.

The startling news that cigarettes and lung cancer were linked caused an instant international uproar. Newspapers, radio, television, and magazines competed with medical journals for the latest scoop. With the exception of the influenza epidemic of 1918, no disease had ever sprung as fast from obscurity to worldwide consciousness. Few have engendered such enormous controversy.

The Hill and Doll study remains one of the crowning glories of medical statistics. It was the first sophisticated case-control study of any noninfectious disease. And it persuaded Hill and Doll to quit smoking. Despite its dramatic results, their study did not show that smoking cigarettes actually causes lung cancer. No one could say that for sure. Jerome Cornfield, an American government bureaucrat at the National Institutes of Health (NIH), took up the challenge. And with Hill organizing clinical studies in Britain and Cornfield developing their mathematical defense in the United States, the two tackled complementary aspects of the same problem from different sides of the Atlantic.

The two men had totally dissimilar backgrounds. Hill's father was a physician with a knighthood, and one of his ancestors had invented the

postage stamp. Cornfield was the son of Russian Jewish immigrants and had earned a bachelor's degree from New York University in 1933. The federal government, desperate for economic data during the Depression, had hired "bright guys" to replace the clerks who had traditionally compiled statistics on unemployment, national income, housing, agriculture, and industry.<sup>1</sup> Cornfield qualified as a bright guy, so he signed on as a government statistician for \$26.31 a week, \$1,368 a year.

Washington, D.C., was still a segregated, southern city. "The rule of thumb was that, if you were Jewish, you could work for the Department of Labor and, if you were Catholic, you could work for the Department of Commerce," explained Marvin Hoffenberg, a friend of Cornfield's and later a UCLA professor.<sup>2</sup> So Cornfield went to Labor. The U.S. Department of Agriculture ran a so-called Graduate School where mathematically inclined government employees could study statistics, and Cornfield took his only mathematics and statistics courses there.

As Cornfield recalled, "Nobody knew how many unemployed there were, and sampling seemed the way to find out. . . . Statistics had me hooked."<sup>3</sup> Although both Fisher and Neyman lectured on sampling methods at the Graduate School, its director, W. Edwards Deming, was open-minded; he published Thomas Bayes' essay with an introduction by Edward Molina of Bell Laboratories.

Friends referred to Cornfield's tenure at the Department of Labor as his serious and exotic phase. He played a major role in revising the Consumer Price Index and in creating one for occupied Japan after the Second World War. But he was "a different kind of a guy," a friend recalled.<sup>4</sup> Unable to think of any good reason for shaving, he grew a little pointed beard, and with his long gaunt frame and an umbrella over his arm he resembled an elegant diplomat strolling jauntily to work. At a time when few others would, he shared his office with a woman statistician and an African American statistical clerk. Next to his mechanical Marchant desktop calculator he installed a Turkish water pipe and could be seen puffing nonchalantly from its two-foot tube.

Cornfield moved to the federal government's new NIH in 1947. Because infectious diseases were in decline in the United States, NIH epidemiologists were attacking chronic diseases, particularly cancer, heart attacks, and diabetes. To assist them, NIH hired a few people with strong quantitative backgrounds. Only one of them had so much as a master's degree. Biostatistics was a professional backwater, and throughout the 1950s and 1960s NIH em-

ployed only ten or 20 statisticians at any one time. It was this small band that introduced statistical methods to NIH researchers in biology and medicine.

By 1950 most men in the United States smoked, and smoking rates were increasing, especially among women. The favorite brands were unfiltered Camels, Lucky Strikes, Chesterfields, and Philip Morris. When Lorillard Tobacco Company introduced filtered Kents in 1952, the filters contained asbestos, which was not removed until 1957. When 14 studies conducted in five countries showed that lung cancer patients included an alarming percentage of heavy smokers, both Cornfield and his wife quit their 2½-pack-a-day habits.

Cornfield realized that the Hill and Wynder studies did not directly answer the questions physicians and their frightened patients were asking: what is my risk? The studies showed the percentages of smokers among groups of people with and without lung cancer, but they did not say what proportion of smokers and nonsmokers was likely to develop lung cancer.

The surest and most direct way to answer the public's fears was to follow large groups of smokers and nonsmokers for years, prospectively, to see how many of each group developed lung cancer. Unfortunately, studies about the future of large populations require a great deal of money and time, especially for relatively rare problems like lung cancer. That is why Hill and Doll had organized their study as a retrospective one, choosing people who already had lung cancer and asking them about their health histories. Such studies are a relatively quick, cheap way to identify potential causes of a particular disease. As a statistician, however, Cornfield suspected that retrospective studies like Hill and Doll's could also be used to answer the individual's haunting question, What's the chance that I or my loved ones will get this fatal disease?

In 1951 Cornfield used Bayes' rule to help answer the puzzle. As his prior hypothesis he used the incidence of lung cancer in the general population. Then he combined that with NIH's latest information on the prevalence of smoking among patients with and without lung cancer. Bayes' rule provided a firm theoretical link, a bridge, if you will, between the risk of disease in the population at large and the risk of disease in a subgroup, in this case smokers. Cornfield was using Bayes as a philosophy-free mathematical statement, as a step in calculations that would yield useful results. He had not yet embraced Bayes as an all-encompassing philosophy.

Cornfield's paper stunned research epidemiologists. More than anything else, it helped advance the hypothesis that cigarette smoking was a cause of lung cancer. Out of necessity, but without any theoretical justification,

epidemiologists had been using case studies of patients to point to possible causes of problems. Cornfield's paper showed clearly that under certain conditions (that is, when subjects in a study were carefully matched with controls) patients' histories could indeed help measure the strength of the link between a disease and its possible cause. Epidemiologists could estimate disease risk rates by analyzing nonexperimental clinical data gleaned from patient histories. By validating research findings arising from case-control studies, Cornfield made much of modern epidemiology possible. In 1961, for example, case-control studies would help identify the antinausea drug thalidomide as the cause of serious birth defects.

Two massive efforts in England and the United States during the mid-1950s confirmed Cornfield's judgment. Because many people had rejected the findings of their retrospective study, Hill and Doll had decided to take a direct approach and conduct a prospective study. They questioned 40,000 British physicians about their current smoking habits and then followed them for five years to see who got lung cancer. In a parallel U.S. study, E. Cuyler Hammond and Daniel Horn followed 187,783 men aged 50 to 69 in New York State for more than 3½ years. Death rates in both countries were similar: heavy smokers were 22 to 24 times more likely to get lung cancer than nonsmokers and, in another surprise discovery, were 42% and 57% more likely to get, respectively, heart and circulatory diseases. Research also showed that cigarettes were more dangerous than pipes, although the risk declined after smoking stopped.

Surprisingly, neither Fisher nor Neyman could accept research results showing that cigarettes caused lung cancer. Both anti-Bayesians were heavy smokers, and Fisher was a paid consultant to the tobacco industry. But more important, neither found epidemiologic studies convincing. And both were correct in pointing out that tobacco could be associated with cancer without causing it. In 1955 they launched a vigorous counterattack, arguing that only experimental data from strictly controlled laboratory and field experiments could predict future disease rates. The most eminent American medical statistician of the day, Joseph Berkson, of the Mayo Clinic in Rochester, Minnesota, joined the attack; Berkson did not believe cigarettes could cause both cancer and heart disease.

Fisher kept up a barrage of angry attacks, including a book and two articles published in highly prestigious journals, *Nature* and the *British Medical Journal*. According to Doll, Fisher even went so far as to accuse Hill of scientific dishonesty. Over the course of three years Fisher developed two remarkable

hypotheses. The first, believe it or not, was that lung cancer might cause smoking. The second was that a latent genetic factor might give some people hereditary predilections for both smoking and lung cancer. In neither case would smoking cause lung cancer.

Cornfield maintained a running argument with Fisher through the 1950s. Cornfield was already thinking deeply about the standards of evidence needed before observational data could establish cause and effect. Finally, in 1959, he raked Fisher over the coals about smoking with a common-sense, nonmathematical paper that reads like a legal brief. In that seminal paper he and five coauthors systematically addressed every one of Fisher's alternative explanations for the link between cigarette smoking and lung cancer. They hurled one counterargument after another at Fisher's hypothetical genetic factor. If cigarette smokers were nine times more likely than nonsmokers to get lung cancer, Fisher's latent genetic factor must be even larger—though nothing approaching that had ever been seen.

Cornfield dismissed out of hand Fisher's suggestion that cancer might cause smoking: "Since we know of no evidence to support the view that the bronchogenic carcinoma diagnosed after age 50 began before age 18, the median age at which smokers begin smoking, we shall not discuss it further."<sup>5</sup> Cornfield pointed out that Fisher's genetic factor would have to spread rapidly and occur more among cigarette smokers than nonsmokers; cause tumors on mouse skin but not on human lungs; weaken with age after a smoker quit; and be more likely in men than women, 60 times more prevalent among two-pack-a-day smokers, and different in pipe and cigar smokers. Yet none of these phenomena had ever been observed.

Fisher wound up looking ridiculous. As Cornfield coolly noted, "A point is reached . . . when a continuously modified hypothesis becomes difficult to entertain seriously."<sup>6</sup> Scientists who can find only one viable explanation for associations in their data have probably found its causal agent. The existence of possible alternative explanations indicates that the cause has probably not yet been found. Cornfield was laying out the road map for future smoking and lung cancer research.

By now, Cornfield the history major had become the most influential biomedical statistician in the United States. When the U.S. surgeon general concluded in 1964 that "cigarette smoking is causally related to lung cancer in men," he cited Cornfield's work.<sup>7</sup> Nonexperimental studies had helped identify an association between smoking and lung cancer. With the help of Bayes' rule—what Laplace had called "the probability of causes and future

events, derived from past events”—Cornfield provided the theoretical justification for using case-control studies to estimate the strength of links between exposure and disease. Today, thanks to Cornfield, case-control studies are the primary tool epidemiologists use to identify likely causes of chronic diseases.

Over his career, Cornfield would become involved in every major public health problem of the day. Most of them, including smoking, the safety of polio vaccines, and the efficacy of diabetes treatments, were fiercely controversial.

To calm the statistics phobia of physicians and epidemiologists, Cornfield developed an easygoing bedside manner. Abandoning his serious phase, he cultivated an infectious laugh and an irrepressible air of informality. By mixing humor into conversations, telling stories, and laughing heartily he inspired tremendous confidence. Even his gait and prose became sprightly. Soon every biomedical scientist with a committee and a controversy wanted Cornfield on board. By pointing out common elements that everyone shared, he could unify the most disparate group. After one particularly onerous series of meetings and reports, a committee member asked him, “Did you get my last letter about sample size?” There was a pause, and Cornfield grinned and said, “Christ, I hope so.” When the committee finally produced its massive procedural manual, Cornfield waved it over his head, declaring, “You know, say what you will about the Ten Commandments, you must come back to the pleasant fact that there were only ten of them.”

Cornfield typically rose at 5 a.m. to write and make paper-and-pencil calculations. He came up with clever approximations and computational tricks, much as Laplace had done. He visualized particularly difficult distributional functions by carving them out of a bar of soap. To collaborate with biochemists, he studied basic biology. And although he was a brilliant speaker, he never prepared a talk until the night before. He procrastinated even the day before he was scheduled to speak at 8:30 a.m. at Yale University about the contentious Salk poliomyelitis vaccine tests. “Don’t worry, Max,” he told a friend. “God will provide.”

Cornfield was a voracious reader but did not own a TV and was blissfully unaware of popular culture. Once a biostatistician who dated Hollywood stars begged him to speed up a morning meeting: “I’ve got to get through because at 12 o’clock I have a luncheon date with Kim Novak.” Puzzled, Cornfield asked, “Kim Novak? Who’s he?”<sup>8</sup> At the time she was Columbia Pictures’ answer to Marilyn Monroe.

Another watershed medical study also occupied Cornfield’s attention

during the 1950s. Death rates from cardiovascular disease had been rising steadily in the United States since 1900. Heart disease had been the nation's leading cause of death since 1921, and strokes the third leading cause since 1938. Yet researchers at the midpoint of the twentieth century were as ignorant of the causes of heart disease and stroke as they had been of lung cancer.

Understanding the causes of deaths from cardiovascular disease would require following a population for many years. A prospective study, however, was more feasible than with lung cancer because heart problems were far more common. In 1948 Cornfield helped design the Framingham Heart Study, which has since followed the health of three generations of Framingham, Massachusetts, residents.

As one of the first important studies based on Framingham, Cornfield followed 1,329 adult male residents for a decade. Between 1948 and 1958, 92 of the group experienced myocardial infarction or angina pectoris.

Longitudinal studies like Framingham are designed to investigate a large variety of variables, both singly and jointly, on the risk of developing a disease. Traditionally, epidemiologists studied their data by inspecting—"contemplating" was Cornfield's word—the resulting multiple cross-classification arrays. Three risk factors, each considered at low, medium, and high levels, would produce a tidy 3x3 table of cells, but as the number of variables increased and they were considered singly and jointly, the number of cells to be contemplated quickly became impracticable. A cross-classification study with 10 risk factors at low, medium, and high levels would produce 59,049 cells. To get even 10 patients per cell, the study would need a cohort of 600,000 people, more than the population of Framingham.

Cornfield realized he needed a "more searching form of analysis than simple inspection."<sup>9</sup> He would have to develop a mathematical model for summarizing the observations. He chose Bayes' rule and used cardiovascular death rates as the prior. Framingham gave him data about two groups of people, those who had died of heart disease and those who had not. Within each group he had information about seven risk factors. Calculating Bayes' rule, he got a posterior probability in the form of a logistic regression function, which he then used to identify the four most important risk factors for cardiovascular disease. In addition to age itself, they were cholesterol, cigarette smoking, heart abnormalities, and blood pressure.

Bayes allowed Cornfield to reframe Framingham's data in terms of the probability that people with particular characteristics would get heart disease.

There was no critical level for cholesterol or blood pressure below which people were safe and above which they got the disease. And patients cursed with both high cholesterol and high blood pressure were 23% more at risk for heart attacks than those with low cholesterol and blood pressure rates.

Cornfield's identification in 1962 of the most critical risk factors for cardiovascular disease produced one of the most important public health achievements of the twentieth century: a dramatic drop in death rates from cardiovascular disease. Between 1960 and 1996 they fell 60%, preventing 621,000 fatalities. His report also showed researchers how to use Bayes' rule to analyze several risk factors at a time; his multiple logistic risk function has been called one of epidemiology's greatest methodologies.

To measure the efficacy of a particular therapy, Cornfield used an early multicenter trial at NIH to introduce another Bayesian concept—Harold Jeffreys's relative betting odds. Now known as the Bayes' Factor, it is the probability of the observed data under one hypothesis divided by its probability under another hypothesis.

When Cornfield worked with researchers who used mice to screen for anticancer drugs, the rigidity of frequentist methods struck him like a blow from behind. According to their rules, even if their initial test results disproved their hypothesis, they had to take six more observations before stopping the experiment. Frequentist methods also banned switching a patient to better treatment before a clinical trial was finished. Frequentist experimenters could not monitor interim results during the clinical trial, examine treatment effects on subgroups of patients, or follow leads from the data with further unplanned analyses. When Cornfield discovered that Bayesian methods would let him reject some hypotheses after only two solidly adverse observations, he was converted. He had started out using Bayes' theorem as an enabling tool to solve a particular problem, the way it had been used for cryptography, submarine hunting, and artillery fire during the Second World War. But now he was moving gradually toward making Bayes' theorem the foundation of a broad philosophy for handling information and uncertainties. As he began thinking about Bayes as a philosophy rather than just a tool, he became part of the profound conversion that Jeffreys, Savage, Lindley, and others were also going through in the 1950s and 1960s. While Fisher considered a hypothesis significant if it was unlikely to have occurred by chance, Cornfield declared airily, "If maintenance of [Fisher's] significance level interferes with the release of interim results, all I can say is so much the worse for the significance level."<sup>10</sup>

Interestingly, most of NIH's other statisticians failed to follow their leader into Bayesian fields. Cornfield published scientifically important papers about Bayesian inference in mainstream statistics journals. Nevertheless, when he included Bayesian methods in some of the trials he worked on, their main conclusions were based on frequentism. It would be another 30 years before NIH started using Bayes for clinical trials. Savage thought that many researchers were content to reap the benefits of Bayes' theorem without embracing the method.

Cornfield declared cheerily, however, "Bayes' theorem has come back from the cemetery to which it has been consigned."<sup>11</sup>

In 1967 Cornfield retired from NIH and moved later to George Washington University, where he chaired the statistics department and developed Bayes' rule into a full-scale logical mathematical approach. In one paper he proved to the satisfaction of many Bayesians that, according to frequency rules, any statistical procedure that does not stem from a prior can be improved upon.

Despite his Bayesian conversion, Cornfield was in great demand as a consultant. He advised the U.S. Army on experimental design; the investigating committee critiquing the best-selling Kinsey Report on female sexuality; the U.S. Department of Justice on sampling voting records to reveal bias against black voters; and the State of Pennsylvania after the Three Mile Island nuclear power plant incident.

In 1974 the Bayesian biostatistician with a bachelor's degree in history became president of the American Statistical Association. In his presidential address, the man who used humor and good cheer to reassure physicians about randomized trials, who gave epidemiologists some of their most important methodologies, and who established causes of both lung cancer and heart attacks, asked, "Why should any person of spirit, of ambition, of high intellectual standards, take any pride or receive any real stimulation and satisfaction from serving an auxiliary role [as a statistician] on someone else's problem?" Smiling at his own question, Cornfield continued, "No one has ever claimed that statistics was the queen of the sciences. . . . The best alternative that has occurred to me is 'bedfellow.' Statistics—bedfellow of the sciences—may not be the banner under which we would choose to march in the next academic procession, but it is as close to the mark as I can come."<sup>12</sup>

When Cornfield was diagnosed with pancreatic cancer in 1979, he knew as well as anyone at NIH the disease's appalling six-month survival rate. Nonetheless, he was determined to continue living to the fullest. Despite

serious postoperative complications, his humor remained intact. A friend told him, "Jerry, I'm so glad to see you." Smiling, Cornfield replied, "That's nothing compared to how happy I am to be able to see you."<sup>13</sup> As he was dying he said to his two daughters, "You spend your whole life practicing your humor for the times when you really need it."<sup>14</sup>

# there's always a first time

Bayes' military successes were still Cold War secrets when Jimmie Savage visited the glamorous new RAND Corporation in the summer of 1957 and encouraged two young men to calculate a life-and-death problem: the probability that a thermonuclear bomb might explode by mistake.

RAND was the quintessential Cold War think tank. Gen. Curtis E. LeMay, the commander of the Strategic Air Command (SAC), had helped start it in Santa Monica, California, 10 years earlier as "a gimmick" to cajole top scientists into applying operations research to long-range air warfare.<sup>1</sup> But RAND, an acronym for Research ANd Development, considered itself a "university without students" and its 1,000-odd employees "defense intellectuals." Their mission was to use mathematics, statistics, and computers to solve military problems, pioneer decision making under conditions of uncertainty, and save the United States from Soviet attack. The U.S. Air Force, which funded RAND, gave its researchers carte blanche to choose the problems they wanted to investigate. But since President Eisenhower's "New Look" military policy depended on early nuclear bombing ("massive retaliation") as the cheapest way to respond to a Soviet attack, RAND's top issues were nuclear strategy, surviving nuclear attack, and response options. Because SAC bombers had a monopoly on transporting America's nuclear arsenal and General LeMay sat at the pinnacle of the world's military might, RAND's voice was often influential.

By the time Savage visited Santa Monica that summer RAND reports had already challenged some of SAC's sacred cows. To drop nuclear weapons on Soviet targets, macho air force pilots wanted to fly the new B-52 Strato-

fortress jets; RAND recommended fleets of cheaper conventional planes. RAND had also described SAC's overseas bases for manned bombers as sitting ducks for Soviet attack. A year after Savage's visit, RAND would challenge Cold War dogma by arguing that victors typically fare better with negotiated settlements than with unconditional surrenders. RAND would even urge counterbalancing LeMay's B-52s with the navy's submarine-based missiles. In retaliation, SAC would almost break off relations with RAND on several occasions between Savage's visit in 1957 and 1961.

Circulating gregariously among RAND researchers that summer, Savage met Fred Charles Iklé, a young Swiss-born demographer who had studied the sociological effects of nuclear bombing on urban populations. At 33, Iklé was seven years younger than Savage and had earned a Ph.D. in 1950 from the University of Chicago, where Savage was teaching. Seeking a wide-open field that no one else at RAND was studying, Iklé chose nuclear catastrophes that an Anglo-American nuclear arsenal would not deter: those caused by accident or by someone mentally deranged. Referring to massive retaliation a few years later, Iklé would declare, "Our method of preventing nuclear war rests on a form of warfare universally condemned since the Dark Ages—the mass killing of hostages."<sup>2</sup> With SAC poised to expand its bomb-bearing flights, Iklé and Savage talked about assessing their impact on nuclear accidents. Eventually the issue came around to the question, what was the probability of an accidental H-bomb explosion?

After a summer of conversations, Savage was preparing to return to academia when Albert Madansky, a 23-year-old Ph.D. graduate of Savage's statistics department, arrived at RAND. Madansky had financed his graduate studies by working part-time for Arthur Bailey, the Bayesian theorist in the insurance industry. Until Bailey's death, he had considered an actuarial career. Savage, who had published his book *Foundations of Statistics* but had not yet embraced Bayes' rule, talked over the H-bomb problem with Madansky without considering it in Bayesian terms. As he left Santa Monica, he handed over the H-bomb study to Madansky but left the young man to do as he saw fit. Madansky would come up with a Bayesian approach on his own.

Because RAND's report on the project would eventually be classified, Madansky could not talk about his work for 41 years. But when Savage returned to Chicago he lectured openly about the fundamental statistical issues involved. Bayes' rule was emerging in fits and starts from the secrecy of the Second World War and the Cold War.

The H-bomb problem facing Madansky was politically and statistically

tricky. No atomic or hydrogen weapon had ever exploded accidentally. In the 12 years since the United States had dropped atomic bombs on Japan in August 1945, nuclear bombs had been detonated, but always deliberately, as part of a weapons test. Barring accidents, the nation's leaders believed that their stocks of nuclear weapons deterred all chance of thermonuclear war—and that accidents could not occur in the future because none had occurred in the past. Nonetheless, the question remained: could the impossible happen?

According to more than a century of conventional statistics, the impossible could not be calculated. Jakob Bernoulli had decreed in 1713 that highly improbable events do not happen. David Hume agreed, arguing that because the sun had risen thousands of times in the past it would continue to do so. It was Thomas Bayes' friend and editor, Richard Price, who took the contrary view that the highly improbable could still occur. During the nineteenth and early twentieth centuries, Antoine-Augustin Cournot concluded that the probability of a physically impossible event was infinitely small and thus the event would never occur. Andrei Kolmogorov slightly refined "never" by saying that if the probability of an event is very small, we can be practically certain that the event will not happen in the very next trial.

Fisher was no help either. He argued that probability is a simple relative frequency in an infinitely large population; until a nuclear bomb accident occurred, he had no way of judging its future probability. Mercifully, Madansky did not have an infinitely large population of H-bomb accidents, and further experimentation was out of the question. Fisher's approach left him with the banal observation that zero accidents had occurred and that the probability of a future accident was also zero.

Madansky concluded, "As long as you are set that the probability is going to be zero, then nothing's going to change your mind. If you have decided that the sun rises each morning because it has always done so in the past, nothing is going to change your mind except one morning when the sun fails to appear."<sup>3</sup>

He did not buy the argument that an accident could never occur simply because none had happened in the past. First, the political and military establishment's assumption that a well-stocked American arsenal of nuclear weapons would deter war rested on increasingly shaky grounds. In the six years between 1949 and 1955 the Soviets had exploded their first atomic bomb, the United States had detonated the world's first hydrogen bomb, and Britain had tested both an atomic and a hydrogen bomb. The USSR had launched the first man-made satellite into orbit around Earth in 1957. Meanwhile, the

United States trained countries in the North Atlantic Treaty Organization (NATO) to fire nuclear weapons and supplied Britain, Italy, and Turkey with nuclear missiles. By the time the Anglo-American Agreement for Co-operation on the Uses of Atomic Energy for Mutual Defense Purposes was signed in 1958, all hope of preventing the spread of nuclear weapons had evaporated. France tested its first atomic bomb in 1960.

In addition to the rapid expansion of nuclear weapons, Madansky had 16 top-secret reasons for doubting that the probability of a future accident was zero. A classified list detailed 16 of “the more dramatic incidents” involving nuclear weapons between 1950 and 1958.<sup>4</sup> They included accidental drops, jettisons, aircraft crashes, and testing errors. Incidents had occurred off British Columbia and in California, New Mexico, Ohio, Florida, Georgia, South Carolina, and overseas. RAND’s list omitted accidents that had not attracted public attention.

An atomic or hydrogen bomb consists of a small capsule, or “pit,” of uranium or plutonium inside a case covered with powerful conventional explosives. Only if these high explosives blow up at the same instant can the uranium or plutonium capsule be sufficiently compressed on all sides to trigger a nuclear blast. In a few cases these conventional explosives had detonated, generally on impact in a plane crash. Because the capsule of nuclear material had not been installed inside the weapons, however, there had been no nuclear accidents. That fact had convinced SAC that its procedures were sound and that no nuclear accident would occur.

Still, scores of people had died when the high explosives in unarmed nuclear weapons blew up. The year 1950 was a banner year for accidents. On April 11, 1950, 13 people died near Kirtland Air Force Base, outside Albuquerque, New Mexico, when a B-29 crashed into a mountain; flames from the high explosives were visible for 15 miles. On July 13, 16 were killed when a B-50 nose-dived near Lebanon, Ohio. Nineteen people, including Gen. Robert F. Travis, died when a B-29 with mechanical problems crash-landed in California on August 5, injuring 60 people in a nearby trailer camp. Also that year, two bombs without nuclear capsules were jettisoned and abandoned in deep water in the Pacific Ocean off British Columbia and in an unnamed ocean outside the United States.

Newspaper reporting was meager until an improperly closed lock in a B-47’s bomb bay opened in 1958 and a “relatively harmless” bomb fell into Walter Gregg’s garden in Mars Bluff, South Carolina.<sup>5</sup> The conventional explosives detonated on impact, digging a crater 30 feet deep and 50 to 70 feet

across, destroying Gregg's house, damaging nearby buildings, and killing several chickens. No people died, but news coverage was extensive; typically reporters said that a "TNT triggering device" had exploded. RAND noted disapprovingly that a *Time* magazine article was "astonishingly accurate."<sup>6</sup> Congress, Britain's Labour Party, and Radio Moscow complained.

The air force paid the Greggs \$54,000, and all B-47 and B-52 flights carrying nuclear weapons were suspended until new safety measures could be introduced. SAC also established a new policy: nuclear bombs were to be jettisoned on purpose only into oceans or designated "water masses. . . . Hence only uncontrolled drops may attract public attention in the future."

As the press became increasingly suspicious of the accidents, Iklé and another RAND researcher, Albert Wohlstetter, became concerned. Iklé recommended that the government remain silent about the presence of nuclear weapons in aircraft crashes. As Iklé and Madansky worked on their study, an accident occurred that could have provoked an international scandal. A wheel casting on a B-47 failed at the air force's fueling base in Sidi Slimane, French Morocco. High explosives detonated and a fire raged for seven hours, destroying the nuclear weapon and capsule aboard.

Given all these accidents involving unarmed nuclear weapons, Madansky felt he could no longer assume, as SAC and frequentist statisticians had, that an accident involving an H-bomb could never occur. Instead, he decided, he needed "another theology, . . . another brand of inference," where the possibility of an accident was not necessarily zero.

Frequentism was no help. "But, but, but," Madansky said later, "if you're willing to admit a shred of disbelief, you can let Bayes' theorem work. . . . Bayes is the only other theology that you can go to. It's just sort of natural for this particular problem. At least that's how I felt back then."<sup>8</sup> As Dennis Lindley had argued, if someone attaches a prior probability of zero to the hypothesis that the moon is made of green cheese, "then whole armies of astronauts coming back bearing green cheese cannot convince him." In that connection, Lindley liked to quote Cromwell's Rule from the Puritan leader's letter to the Church of Scotland in 1650: "I beseech you, in the bowels of Christ, think it possible you may be mistaken."<sup>9</sup> In the spirit of Cromwell's Rule, Madansky adopted Bayes as his "alternate theology."

Many Cold War statisticians knew it well. They were using it to deal with one of their biggest problems, estimating the reliability of the new intercontinental ballistic missiles. "We didn't know how reliable the missiles were," Madansky explained, "and we had a limited amount of test data

to determine it, and so a number of the people working in reliability applied Bayesian methods. The entire field, from North American Rockwell, Thompson Ramo Wooldridge, Aerospace, and others, was involved. I'm sure Bayesian ideas floated around them too. We all knew about it. . . . It was just a natural thing to do."<sup>10</sup>

Madansky immediately set out to measure how much credence could be given to the belief that no unauthorized detonations would occur in the future. He started with "statistically speaking, this simple commonsense idea based on the notion that there is an *a priori* distribution of the probability of an accident in a given opportunity, which is not all concentrated at zero."<sup>11</sup> The decision to incorporate a whisper of doubt into his prior was important. Once a Bayesian considered the possibility that even one small accident might have occurred in the past 10,000 opportunities, the probability of an accident-free future dropped significantly.

Politically and mathematically, Madansky faced an extremely difficult problem. As a young civilian challenging one of the military's fundamental beliefs about the Cold War, he would have to convince decision makers that, although nothing catastrophic had occurred, something might in future. He would have to explain the Bayesian process to nonspecialists. And because the military was often suspicious of civilians making unwarranted suggestions, he would have to make as few initial assumptions as possible. In addition, there was the statistical problem involving the small number of accidents and, fortunately, no Armageddons.

Not wanting to stick his neck out, Madansky decided to make his prior odds considerably weaker than 50–50. "I tried to figure out how to avoid making any specifications about what the prior should be."<sup>12</sup> To refine his minimally informative prior, he added another common-sense notion: the probability of an accident-free future depends on the length of the accident-free past and the number of future accident opportunities. Madansky had no direct evidence because there had never been a nuclear accident. But the military had plenty of indirect data, and he began using it to modify his prior.

He knew that the military was already developing plans to greatly increase the number of flights carrying nuclear weapons. SAC was planning a system of 1,800 bombers capable of carrying nuclear weapons; approximately 15% of them would be in the air at all times, armed and ready for attack. At that time, SAC's B-52 jet-powered Stratofortress bombers carried up to four nuclear bombs, each with an explosive power between 1 million and 24 million tons of TNT, or up to 1,850 Hiroshima bombs. The United States

was also planning to outfit intercontinental ballistic missiles with hydrogen warheads, accelerate the production of intermediate range ballistic missiles, and negotiate with NATO countries for launching rights and military bases. The military would soon be dealing with shorter alarm times, increased alertness, and more decentralized weaponry, all factors that could increase the likelihood of a catastrophe.

Madansky calculated the number of “accident opportunities” based on the number of weapons, their longevity, and the number of times they were aboard planes or handled in storage.<sup>13</sup> Accident opportunities corresponded to flipping coins and throwing dice. Counting them proved to be an important innovation.

“A probability that is very small for a single operation, say one in a million, can become significant if this operation will occur 10,000 times in the next five years,” Madansky wrote.<sup>14</sup> The military’s own evidence indicated that “a certain number of aircraft crashes” was inevitable. According to the air force, a B-52 jet, the plane carrying SAC’s bombs, would average 5 major accidents per 100,000 flying hours. Roughly 3 nuclear bombs were dropped accidentally or jettisoned on purpose per 1,000 flights that carried these weapons. In that 80% of aircraft crashes occurred within 3 miles of an air force base, the likelihood of public exposure was growing. And so it went. None of these studies involved a nuclear explosion, but to a Bayesian they suggested ominous possibilities.

Computationally, Madansky was confident that RAND’s two powerful computers, a 700 series IBM and the Johnniac, designed by and named for John von Neumann, could handle the job. But he hoped to avoid using them by solving the problem with pencil and paper.

Given the power and availability of computers in the 1950s, many Bayesians were searching for ways to make calculations manageable. Madansky latched onto the fact that many types of priors and posteriors share the same probability curves. Bailey had used the same technique in the late 1940s, and later it would be known as Howard Raiffa and Robert Schlaifer’s conjugate priors. When Madansky read their book describing them, he was pleased to know that his prior had a name and rationale: “I was just doing it ad hoc.”<sup>15</sup>

Using his tractable prior, classified military data, and informed guesswork, Madansky came to a startling conclusion. It was highly probable that SAC’s expanded airborne alert system would be accompanied by 19 “conspicuous” weapon accidents each year.

Madansky wrote an elementary summary that high-level military deci-

sion makers could understand and inserted it into RAND's final report, "On the Risk of an Accidental or Unauthorized Nuclear Detonation. RM-2251 U.S. Air Force Project Rand," dated October 15, 1958. The report listed Iklé as its primary author in collaboration with Madansky and a psychiatrist, Gerald J. Aronson. Until then, many RAND reports had been freely published, but air force censors were clamping down on its think tank, and this report was classified, its initial readership limited to a select few. It was finally released more than 41 years later, on May 9, 2000, with numerous passages blacked out.

In view of what would happen a few years later in Spain, much of the report seems prescient. Madansky could not predict when or where the accident would occur, but he was sure of two things. The likelihood of an accident was rising, and it was in the military's interest to make its nuclear arsenal safer. Given the media's increasingly savvy coverage of accidents involving nuclear weaponry, Iklé foresaw Soviet propaganda, citizens' campaigns to limit the use of nuclear devices, and foreign powers demanding an end to American bases on their soil. Who knew, but the British Labour Party might even win an election or NATO might crumble?

As a result, Iklé and Madansky advocated safety features that included requiring at least two people to arm a nuclear weapon; electrifying arming switches to jolt anyone who touched them; arming weapons only over enemy territory; installing combination locks inside warheads; preventing the release of radioactive material in accidental fires of high-energy missile fuels; and switching the nuclear matter inside weapons from plutonium to uranium because the latter would contaminate a smaller area. The report also recommended placing reassuring articles in scientific journals to publicize research indicating that plutonium emissions were less dangerous to humans than had been thought; the source of the research was to be disguised.

Later, after SAC implemented its airborne alert program and began keeping significant numbers of nuclear-armed planes aloft at all times, Iklé and Madansky followed up with a more pointed and less mathematical summary about accident rates. As of 2010 this internal document, meant only for internal circulation, was still classified. In an appendix to the first report, Iklé and Aronson, the psychiatrist who consulted at RAND, tackled the topic of mental illness among military personnel in charge of nuclear bombs. Such concerns were widespread at the time. Aronson believed that men who worked near the bombs should be psychologically tested by being confined in a chamber by themselves, deprived of sleep and sensory stimuli for several hours, and perhaps dosed with hallucinogens such as LSD. He predicted "only

between one-third and one-quarter of 'normal' volunteer-subjects would be able to withstand [it] for more than several hours."<sup>16</sup> It was later learned that the Central Intelligence Agency financed a variety of LSD (lysergic acid diethylamide) experiments on people without their knowledge or consent during the 1950s, although the practice violated ethical standards even then.

After the report was issued, Iklé, his knees shaking, went to brief "a sizable audience of Air Force generals."<sup>17</sup> RAND researchers assumed that General LeMay would scorn their conclusions. LeMay had directed the fire-bombing of Japanese cities during the Second World War, his ghostwritten autobiography would propose bombing the Vietnamese "back into the Stone Age" in the mid-1960s, and he was the model for "Buck" Turgidson, the insanely warlike, cigar-chomping general played by George C. Scott in the film *Dr. Strangelove*. Iklé described LeMay's attitude about nuclear weapons as one of "injudicious pugnacity."<sup>18</sup>

But the general surprised him. The day after RAND presented its report in Washington, LeMay asked for a copy. Iklé said LeMay later issued a "blizzard" of orders, among others calling for the two-man rule and coded locks. The army and navy followed suit. Iklé put LeMay's response "in the 'success' column of my life's ledger."<sup>19</sup>

According to most reports, however, few of the coded locks were actually installed on nuclear weapons until John F. Kennedy became president. Four days after Kennedy's inauguration, a SAC B-52 disintegrated in midflight. One of its two 24-megaton hydrogen bombs smashed into a swamp near Goldsboro, North Carolina, and a large chunk of enriched uranium sank more than 50 feet, where it presumably remains to this day. Analysis showed that only one of the bomb's six safety devices had functioned properly. JFK was told of scores of nuclear weapon accidents—according to *Newsweek*, more than 60 since the Second World War. From then on, the Kennedy administration vigorously pursued nuclear weapon safety and added coded locks to nuclear weapons.

Iklé went on to become a leading hard-line specialist in military and foreign policy and would be awarded two Distinguished Public Service Medals, the Department of Defense's highest civilian award. Madansky became a professor at the University of Chicago, where he developed a reputation as a neutral pragmatist in the battles between Bayesians and frequentists. RAND gradually weaned itself from air force funding by diversifying into social welfare research.

The world can be thankful that Madansky's Bayesian statistics forced

the military to tighten safety measures. A number of false alerts suggestive of Soviet nuclear attacks were identified correctly before SAC could launch a counterattack. Phenomena causing false alerts included the aurora borealis, a rising moon, space debris, false U.S. radar signals, errors in the use of computers (such as a mistaken Pentagon warning of incoming Soviet missiles in 1980), routine Soviet maintenance procedures after the Chernobyl accident, a Norwegian weather research missile, and “more hidden problems of unauthorized acts.”

# 46,656 varieties

In sharp contrast to the super secrecy of Madansky's H-bomb report, the schism between entrenched frequentists and upstart Bayesians was getting downright noisy. As usual, the bone of contention was the subjectivity of Thomas Bayes' pesky prior. The idea of importing knowledge that did not originate in the statistical data at hand was anathema to the anti-Bayesian duo Fisher and Neyman. Since they were making conclusions and predictions about data without using prior odds, Bayesian theoreticians on the defensive struggled to avoid priors altogether.

Bayesian theories mushroomed in glorious profusion during the 1960s, and Jack Good claimed he counted "at least 46656 different interpretations," far more than the world had statisticians.<sup>1</sup> Versions included subjective, personalist, objective, empirical Bayes (EB for short), semi-EB, semi-Bayes, epistemic, intuitionist, logical, fuzzy, hierarchical, pseudo, quasi, compound, parametric, nonparametric, hyperparametric, and non-hyperparametric Bayes. Many of these varieties attracted only their own creators, and some modern statisticians contend that hairsplitting produced little pathbreaking Bayesian theory. When asked how to differentiate one Bayesian from another, a biostatistician cracked, "Ye shall know them by their posteriors."

Almost unnoticed in the hoopla, the old term "inverse probability" was disappearing, and a modern term, "Bayesian inference," was taking its place. As the English language swept postwar statistical circles, articles by British theorists began to look more important than Laplace's French ones. "Much that's been written about the history of probability has been distorted by this English-centric point of view," says Glenn Shafer of Rutgers University.<sup>2</sup>

More than language may have been involved. In 2008, when the Englishman Dennis Lindley was 85 years old, he said he was now almost convinced that Laplace was more important than Thomas Bayes: "Bayes solved one narrow problem; Laplace solved many, even in probability. . . . My ignorance of the Frenchman's work may be cultural, since he did not figure prominently in my mathematical education." Then he added with characteristic honesty, "But I am biased: the French let us down during WW2 and then there was the ghastly de Gaulle."<sup>3</sup>

In England the ferment over Bayes' rule extended even into Fisher's family. His son-in-law George E. P. Box was the young chemist who had to say he was transporting a horse in order to consult Fisher during the Second World War. Like Fisher, Box came to believe that statistics should be more closely entwined with science than with mathematics. This view was reinforced by Box's later work for the chemical giant ICI in Britain and for the quality control movement with W. Edwards Deming and the Japanese auto industry.

When Box organized a statistics department at the University of Wisconsin in 1960, he taught for the first time, a class called Foundations of Statistics. "Week after week," he said, "I prepared my notes very carefully. But the more I did, the more I became convinced that the standard stuff I'd studied under Egon Pearson was wrong. So gradually my course became more and more Bayesian. . . . People used to make fun of it and say it was all nonsense."<sup>4</sup>

Helping scientists with scanty data, Box found that traditional statistics produced messy, unsatisfactory solutions. Still, frequentism worked well for special cases where data fell into bell-shaped probability curves and middle-values were assumed to be averages. So, as Box said, "Comparing averages looked right to me until Stein."<sup>5</sup>

Stein's Paradox called those averages into question. Charles Stein, a theoretical statistician, had been thinking about something that looked quite simple: estimating a mean. Statisticians do not concern themselves with individuals; their bread and butter is a midvalue summarizing large amounts of information. The centuries-old question was which midvalue works best for a particular problem. In the course of his investigation, Stein discovered a method that, ironically, produced more accurate predictions than simple averages did. Statisticians called it Stein's Paradox. Stein called it shrinkage. As a frequency-based theorist, he studiously avoided discussing its relationship with Bayes.

Stein's Paradox, however, works for comparisons between related statistics: the egg production of different chicken breeds, the batting averages of

various baseball players, or the workers' compensation exposure of roofing companies. Traditionally, for example, farmers comparing the egg production of five chicken breeds would average the egg yields of each breed separately. But what if a traveling salesman advertised a breed of hens, and each one supposedly laid a million eggs? Because of their prior knowledge of poultry, farmers would laugh him out of town. Bayesians decided that Stein, like the farmers, had weighted his average with a sort of super- or hyperdistribution about chicken-ness, information about egg laying inherent in each breed but never before considered. And intrinsic to poultry farming is the fact that one hen never lays a million eggs.

In like manner, Stein's system used prior information to explain the hitherto mediocre batter who begins a new season hitting a spectacular .400. Stein's Paradox tells fans not to forget their previous knowledge of the sport and the batting averages of other players.

When Stein and Willard D. James simplified the method in 1961, they produced another surprise—the same Bayes-based formula that actuaries had used earlier in the century to price workers' compensation insurance premiums. Whitney's actuarial Credibility theorem used  $x = P + z(p - P)$ , and Stein and James used  $z = \bar{y} + c(y - \bar{y})$ : identical formulas but with different symbols and names. In both cases, data about related quantities were being concentrated, made more credible or shrunk, until they clustered more tightly around the mean. With it, actuaries made more accurate predictions about the future well-being of workers in broad industrial categories. Only Arthur Bailey had recognized the formula's Bayesian roots and realized it would be equally valid for noninsurance situations.

Delighted Bayesians claimed Stein was using the prior context of his numbers to shrink the range of possible answers and make for better predictions. Stein, however, continued to regard Bayes' philosophical framework as a "negative" and subjective priors as "completely inappropriate."<sup>6</sup>

Box, who believed Stein should have admitted his method was Bayesian, immediately thought of other relationships that work the same way. Daily egg yields produced on Monday are related to yields produced on Tuesday, Wednesday, and Thursday. In this case, the link between different items is a time series, and successive observations tend to be correlated with each other just as tomorrow tends to be rather like today. But Box discovered gleefully that analyzing time series with Bayesian methods improved predictions and that without those methods Stein's Paradox did not work for time series. As Box explained, "If someone comes into your office with some numbers and

says, 'Analyze them,' it's reasonable to ask where they come from and how they're linked. The quality that makes numbers comparable has to be taken into account. You can't take numbers out of context."<sup>7</sup>

Frequentists and Bayesians battled for years over Stein's Paradox, in part because neither side seemed to be all right or all wrong. Box, however, was a convinced Bayesian, and he wrote a snappy Christmas party song to the tune of "There's No Business like Show Business." One verse went:

There's no Theorem like Bayes theorem  
Like no theorem we know  
Everything about it is appealing  
Everything about it is a wow  
Let out all that a priori feeling  
You've been concealing right up to now.  
. . . . There's no theorem like Bayes theorem  
Like no theorem we know.<sup>8</sup>

As Bayesian interpretations multiplied like bunnies and cropped up in unlikely places like Stein's Paradox, cracks formed in Fisher's favorite theory, fiducial probability. He had introduced it as an alternative to Bayes' rule during an argument with Karl Pearson in 1930. But in 1958 Lindley showed that when uniform priors are used, Fisher's fiducial probability and Bayesian inference produce identical solutions.

Still another fissure occurred when Allan Birnbaum derived George A. Barnard's likelihood principle from generally accepted frequentist principles and showed that he needed to take into account only observed data, not information that might have arisen from the experiment but had not. Another frequentist complained that Birnbaum was "propos[ing] to push the clock back 45 years, but at least this puts him ahead of the Bayesians, who would like to turn the clock back 150 years."<sup>9</sup> Jimmie Savage, however, praised Birnbaum's work as a "historic occasion."<sup>10</sup>

Savage also damned Fisher's fiducial method for using parts of Bayes while avoiding the opprobrium attached to priors. Savage considered Fisher's theory "a bold attempt to make the Bayesian omelet without breaking the Bayesian eggs."<sup>11</sup> Box thought his father-in-law's fiducial probability was beginning to look like "a sneaky way of doing Bayes."<sup>12</sup>

Yet another disagreement between Bayesians and anti-Bayesians surfaced in 1957, when Lindley, elaborating on a point made by Jeffreys, highlighted a theoretical situation when the two approaches produce diametrically

opposite results. Lindley's Paradox occurs when a precise hypothesis is tested with vast amounts of data. In 1987, Princeton University aeronautics engineering professor Robert G. Jahn conducted a large study which he concluded supported the existence of psychokinetic powers. He reported that a random event generator had produced 104,490,000 trials testing the hypothesis that someone on a couch eight feet away *cannot* influence their results any more than random chance would. Jahn reported that the random event generator produced 18,471 more examples (0.018%) of human influence on his sensitive microelectronic equipment than could be expected with chance alone. Even with a  $p$ -value as small as 0.00015, the frequentist would reject the hypothesis (and conclude in favor of psychokinetic powers) while the same evidence convinces a Bayesian that the hypothesis against spiritualism is almost certainly true.

Six years later, Jimmie Savage, Harold Lindman, and Ward Edwards at the University of Michigan showed that results using Bayes and the frequentist's  $p$ -values could differ by significant amounts even with everyday-sized data samples; for instance, a Bayesian with any sensible prior and a sample of only 20 would get an answer ten times or more larger than the  $p$ -value.

Lindley ran afoul of Fisher's temper when he reviewed Fisher's third book and found "what I thought was a very basic, serious error in it: Namely, that [Fisher's] fiducial probability doesn't obey the rules of probability. He said it did. He was wrong; it didn't, and I produced an example. He was furious with me." A sympathetic colleague warned Lindley that Fisher was livid, but "it was not until the book of Fisher's letters was published that I realized the full force of his fury. He was unreasonable; he should have admitted his mistake. Still, I was a bumptious young man and he was entitled to be a bit angry." Lindley compounded his tactlessness by turning his discovery into a paper. The journal editor agreed that the article had to be published because it was correct, but asked Lindley if he knew what he was doing: "We would have the wrath of Fisher on our heads."<sup>13</sup> For the next eight months Fisher's letters to friends included complaints about Lindley's "rather abusive review."<sup>14</sup>

Bayes frayed Neyman's nerves too. When Neyman organized a symposium in Berkeley in 1960, Lindley read a paper about prior distributions that caused "the only serious, public row that I can recall having had in statistics. Neyman was furious with me in public. I was very worried, but Savage leapt to my defense and dealt with the situation, I think very well."<sup>15</sup>

One day in the mid-1960s Box dared to tackle his irascible father-in-law

about equal priors. Fisher had come to see his new granddaughter, and friends had warned Box he could get only so far with Fisher before he exploded. Walking up the hill to the University of Wisconsin in Madison, Box told the old man, "I'll give them equal probabilities, so if I have five hypotheses, each has a probability of one-fifth."<sup>16</sup>

Fisher responded in a rather annoyed way, as if to say, "This is what I'm going to say and then you're going to shut up." He declared, "Thinking that you don't know and thinking that the probabilities of all the possibilities are equal is not the same thing."<sup>17</sup> That distinction, which Box later agreed with, prevented Fisher from accepting Bayes' rule. Like Neyman, Fisher agreed that if he ever saw a prior he could believe, he would use Bayes-Laplace. And in fact he did. Because he knew the pedigrees of his laboratory animals generations back, he could confidently specify the initial probabilities of particular crosses. For those experiments Fisher did use Bayes' rule. Later, Box said sadly that he divorced Fisher's daughter because she had inherited a temper much like her father's.

A compromise between Bayesian and anti-Bayesian methods began to look attractive. The idea was to estimate the initial probabilities according to their relative frequency and then proceed with the rest of Bayes' rule. Empirical Bayes, as it was called, seemed like a breakthrough. Egon Pearson had made an early stab at it in 1925, Turing had used a variant during the Second World War, Herbert Robbins proposed it in 1955, and when Neyman recommended it, a flurry of publications appeared. Empirical Bayesians had little impact on mainstream statistical theory, however, and almost none on applications until the late 1970s.

At the same time, others tackled one of the practical drawbacks to Bayes: its computational difficulty. Laplace's continuous form of Bayes' rule called for integration of functions. This could be complicated, and as the number of unknowns rose, integration problems became hopelessly difficult given the computational capabilities of the time. Jeffreys, Lindley, and David Wallace were among those who worked on developing asymptotic approximations to make the calculations more manageable.

Amid this mathematical fervor, a few practical types sat down in the 1960s to build the kind of institutional support that frequentists had long enjoyed: annual seminars, journals, funding sources, and textbooks. Morris H. DeGroot wrote the first internationally known text on Bayesian decision theory, the mathematical analysis of decision making. Arnold Zellner at the University of Chicago raised money, founded a conference series, and

began testing standard economics problems one by one, solving them from both Bayesian and non-Bayesian points of view. Thanks to Zellner's influence, Savage's subjective probability would have one of its biggest impacts in economics. The building process took decades. The International Society for Bayesian Analysis and the Bayesian section of the American Statistical Association were not formed until the early 1990s.

The excitement over Bayesian theory extended far beyond statistics and mathematics though. During the 1960s and 1970s, physicians, National Security Agency cryptanalysts, Central Intelligence Agency analysts, and lawyers also began to consider Bayesian applications in their fields.

Physicians began talking about applying Bayes to medical diagnosis in 1959, when Robert S. Ledley of the National Bureau of Standards and Lee B. Lusted of the University of Rochester School of Medicine suggested the idea. They published their article in *Science* because medical journals were uninterested. After reading it, Homer Warner, a pediatric heart surgeon at the Latter-day Saints Hospital and the University of Utah in Salt Lake City, developed in 1961 the first computerized program for diagnosing disease. Basing it on 1,000 children with various congenital heart diseases, Warner showed that Bayes could identify their underlying problems quite accurately. "Old cardiologists just couldn't believe that a computer could do something better than a human," Warner recalled.<sup>18</sup> A few years after Warner introduced his battery of 54 tests, Anthony Gorry and Otto Barnett showed that only seven or eight were needed, as long as they were relevant to the patient's symptoms and given one at a time in proper sequence. Few physicians used the systems, however, and efforts to computerize diagnostics lapsed.

Between 1960 and 1972 the National Security Agency educated cryptanalysts about advanced Bayesian methods with at least six articles in its in-house *NSA Technical Journal*. Originally classified as "Top Secret Umbra," an overall code word for high-level intelligence, they were partially declassified at my request in 2009, although the authorship of three of the six was suppressed. (At least one of those three has the earmarks of a Jack Good article.) In another article, an agency employee named F. T. Leahy quoted the assertion in Van Nostrand's *Scientific Encyclopedia* that Bayes' theorem "has been found to be unscientific, to give rise to various inconsistencies, and to be unnecessary." In fact, Leahy declared in 1960, Bayes is "one of the more important mathematical techniques used by cryptanalysts . . . [and] was used in almost all the successful cryptanalysis at the National Security Agency. . . . It leads to the only correct formulas for solving a large number of our cryptanalytic

problems,” including those involving comparisons among a multiplicity of hypotheses.<sup>19</sup> Still, “only a handful of mathematicians at N.S.A. know about all the ways” Bayes can be used. The articles were presumably intended to remedy the situation.

At the CIA, analysts conducted dozens of experiments with Bayes. The CIA, which must infer information from incomplete or uncertain evidence, had failed at least a dozen times to predict disastrous events during the 1960s and 1970s. Among them were North Vietnam’s intervention in South Vietnam and the petroleum price hike instituted by the Organization of Petroleum Exporting Countries (OPEC) in 1973. Agency analysts typically made one prediction and left it at that; they ignored unlikely but potentially catastrophic possibilities and failed to update their initial predictions with new evidence. Although the CIA concluded that Bayes-based analyses were more insightful, they were judged too slow. The experiments were abandoned for lack of computer power.

The legal profession reacted differently. After several suggestions that Bayes might be useful in assessing legal evidence, Professor Laurence H. Tribe of Harvard Law School published a blistering and influential article about the method in 1971. Drawing on his bachelor’s degree in mathematics, Tribe condemned Bayes and other “mathematical or pseudo-mathematical devices” because they are able to “distort—and, in some instances, to destroy—important values” by “enshrining the [legal] process in mathematical obscurity.”<sup>20</sup> After that, many courtroom doors slammed shut on Bayes.

The extraordinary fact about the glorious Bayesian revival of the 1950s and 1960s is how few people in any field publicly applied Bayesian theory to real-world problems. As a result, much of the speculation about Bayes’ rule was moot. Until they could prove in public that their method was superior, Bayesians were stymied.